

* Probability distribution functions:-

describe how the probabilities are distributed over the values of a random variable.

Ex:- Ages = { --- } $\Rightarrow$ continuous random variable



2 types of Probability distribution functions:-

- ① Probability mass functions (PMF): used for discrete random variables.
- ② Probability density functions (PDF): used for continuous random variables.

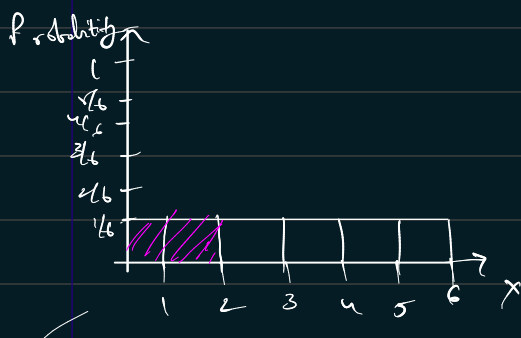
① Prob. mass function (PMF):-

$\rightarrow$ for discrete random variable

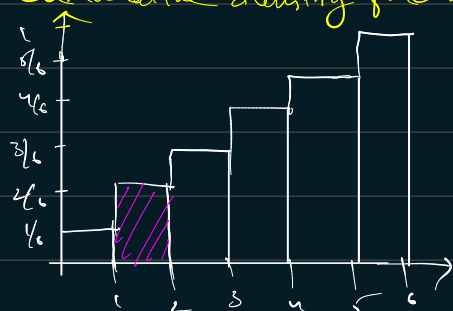
Ex:- Rolling a dice {1, 2, 3, 4, 5, 6}

- Fair dice

$$P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = 1/6$$



Cumulative density function (CDF)



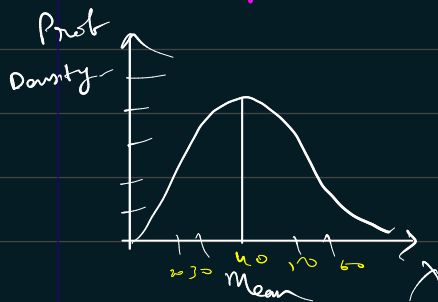
Ex:-

$$\begin{aligned} P(X \leq 2) &= P(X=1) + P(X=2) \\ &= 1/6 + 1/6 = 2/6 \end{aligned}$$

② Prob. Density function (PDF)

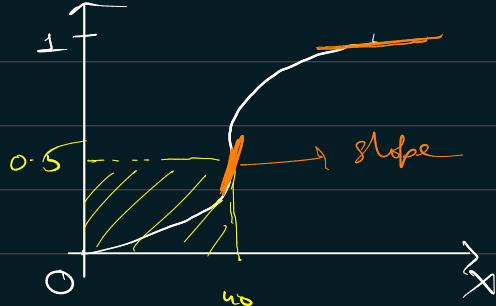
→ For Continuous random variable,

$$X = \text{Ages} = \{ \dots \}$$



Cumulative density function

Cumulative Prob



$$\text{slope} = \frac{x_2 - x_1}{y_2 - y_1}$$

$$P(X \leq u_0) = 0.5 = 50\%$$

Probability density:-

Gradient of cumulative density function

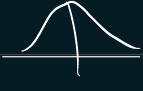
Properties of prob. dist. function:-

1. Non negativity $f(x \geq 0) \quad \forall x$

2. The total area under the PDF curve is equal to 1.

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

* Types of Prob. Distributions:-

- ① Bernoulli dist. $\rightarrow$ (when outcomes are binary) $\Rightarrow$ discrete random variable
 Ex:- Tossing a coin PMF
- ② Binomial dist. $\Rightarrow$ PMF (discrete random variable)
- ③ Normal / Gaussian dist. $\Rightarrow$ PDF 
- ④ Poisson dist. $\Rightarrow$ PMF
- ⑤ Log normal dist. $\Rightarrow$ PDF
- ⑥ uniform dist. $\Rightarrow$ PMF

Dataset

Ex:- House price prediction

Size of house	No. of rooms	Location	# floors	Seaside	Price
$\downarrow$	$\downarrow$		$\downarrow$	$\downarrow$	$\downarrow$
Continuous	Discrete		Discrete	$\{0, 1\}$	Continuous
PDF	PMF		PMF	PMF	PDF

① Bernoulli distribution

-> simplest discrete probability distribution (PMF)

-> represents the probability distribution of a random variable that has exactly two possible outcomes: success (with probability p) and failure (with probability $1-p$).

-> used to model binary outcomes, such as coin flip or yes/no questions.

Ex:- Tossing a coin

$$P(H) = 0.5 = p$$

$$P(T) = 1 - p = 1 - 0.5 = 0.5$$

Person will pass or fail.

$$P(X = \text{Pass}) = 0.4$$

$$P(X = \text{Fail}) = 1 - p = 0.6$$

Parameters:-

$$0 \leq p \leq 1$$

$$P(\text{Success}) \Rightarrow k=1$$

$$q = 1 - p$$

$$P(\text{fail}) \Rightarrow k=0$$

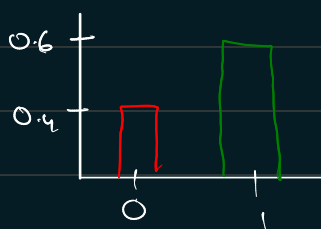
$$k = \{0, 1\} \Rightarrow 2 \text{ outcomes}$$

Ex:- Company launched a new product.

Reviews

$$\text{Good} - 60\% \Rightarrow p$$

$$\text{bad} - 40\% \Rightarrow q = 1 - p$$



$$PMF = p^k \cdot (1-p)^{1-k}$$

$$\text{for } k=1$$

$$P(k=1) = p^1 \cdot (1-p)^0 = p$$

$$\text{for } k=0$$

$$P(k=0) = p^0 \cdot (1-p)^1 = (1-p) = q$$

Bernoulli distribution	
Probability mass function	
Three examples of Bernoulli distribution:	
$P(x=0) = 0.2$ and $P(x=1) = 0.8$ $P(x=0) = 0.8$ and $P(x=1) = 0.2$ $P(x=0) = 0.5$ and $P(x=1) = 0.5$	
Parameters	$0 \leq p \leq 1$ $q = 1 - p$
Support	$k \in \{0, 1\}$
PMF	$\begin{cases} q = 1 - p & \text{if } k = 0 \\ p & \text{if } k = 1 \end{cases}$
CDF	$\begin{cases} 0 & \text{if } k < 0 \\ 1 - p & \text{if } 0 \leq k < 1 \\ 1 & \text{if } k \geq 1 \end{cases}$
Mean	p
Median	$\begin{cases} 0 & \text{if } p < 1/2 \\ [0, 1] & \text{if } p = 1/2 \\ 1 & \text{if } p > 1/2 \end{cases}$
Mode	$\begin{cases} 0 & \text{if } p < 1/2 \\ 0, 1 & \text{if } p = 1/2 \\ 1 & \text{if } p > 1/2 \end{cases}$
Variance	$p(1 - p) = pq$
MAD	$2p(1 - p) = 2pq$
Skewness	$\frac{q - p}{\sqrt{pq}}$
Excess kurtosis	$\frac{1 - 6pq}{pq}$
Entropy	$-q \ln q - p \ln p$
MGF	$q + pe^t$
CF	$q + pe^{it}$
PGF	$q + pz$
Fisher information	$\frac{1}{pq}$

(Simplified)

$$PMF = \begin{cases} q = 1-p & \text{if } k=0 \\ p & \text{if } k=1 \end{cases}$$

- Mean of Bernouli Dist.

$$E(X) = \sum_{k=0}^K k P(k)$$

$E \rightarrow$ Expected

$$= 0(0.4) + 1(0.6)$$

$$= 0.6 \Rightarrow \underline{p}$$

$$\text{Mean} = p$$

- Median of Bernouli dist.

$$\text{Median} = \begin{cases} 0 & \text{if } p < 1/2 \\ [0, 1] & \text{if } p = 1/2 \\ 1 & \text{if } p > 1/2 \end{cases}$$

- Mode of Bernouli dist

$$p \quad \text{if } p > q \quad (\text{same as median})$$

$$q \quad \text{if } q > p$$

- Variance

$$\sigma^2 = 0.4 \cdot (0 - 0.6)^2 + 0.6 \cdot (1 - 0.6)^2$$

$$= 0.4 \cdot (0.36) + 0.6 \cdot (0.16)$$

$$= 0.24 \Rightarrow \frac{P(k=0) \cdot P(k=1)}{q \cdot p}$$

$$\sigma^2 = p \cdot q$$

$$sd = \sigma = \sqrt{p \cdot q}$$

② Binomial Distribution:-

-> is the discrete probability distribution of the number of successes in a sequence of n independent experiments, each asking a yes-no question, and each with its own Boolean-valued outcome: success (with probability p) or failure (with probability $q = 1 - p$).

-> A single success/failure experiment is also called a Bernoulli trial or Bernoulli experiment, and a sequence of outcomes is called a Bernoulli process; for a single trial, i.e., $n = 1$, the binomial distribution is a Bernoulli distribution. The binomial distribution is the basis for the binomial test of statistical significance.

1 trial $\rightarrow$ Bernoulli dist

n trials $\rightarrow$ Binomial dist

Notation: $B(n, p)$

Parameters:

$n \in \{0, 1, 2, \dots\} \Rightarrow$ no. of trials / experiments.

$p \in [0, 1] \rightarrow$ success probability for each trial

$q = 1 - p$.

Support parameter:

$k \in \{0, 1, \dots, n\} \rightarrow$ no. of successes

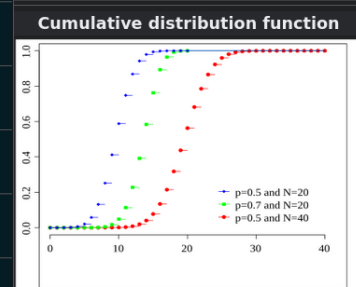
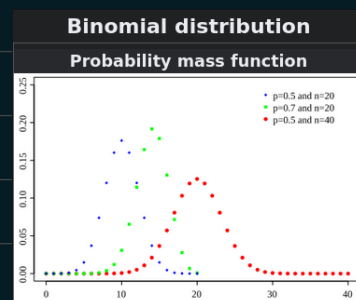
$$PMF = P(k, n, p) = {}^nC_k p^k (1-p)^{n-k}$$

$${}^nC_k = \frac{n!}{(n-k)! k!} \rightarrow \text{Binomial coefficient}$$

Mean: np

Variance: npq

SD: $\sqrt{npq}$



Notation	$B(n, p)$
Parameters	$n \in \{0, 1, 2, \dots\}$ - number of trials $p \in [0, 1]$ - success probability for each trial $q = 1 - p$
Support	$k \in \{0, 1, \dots, n\}$ - number of successes
PMF	$\binom{n}{k} p^k q^{n-k}$
CDF	$I_q(n - \lfloor k \rfloor, 1 + \lfloor k \rfloor)$ (the regularized incomplete beta function)
Mean	np
Median	$\lfloor np \rfloor$ or $\lceil np \rceil$
Mode	$\lfloor (n+1)p \rfloor$ or $\lceil (n+1)p \rceil - 1$
Variance	$npq = np(1-p)$
Skewness	$\frac{q-p}{\sqrt{npq}}$
Excess kurtosis	$\frac{1-6pq}{npq}$
Entropy	$\frac{1}{2} \log_2(2\pi e npq) + O\left(\frac{1}{n}\right)$ in shannons. For nats, use the natural log in the log.
MGF	$(q + pe^t)^n$
CF	$(q + pe^{it})^n$
PGF	$G(z) = [q + pz]^n$
Fisher information	$g_n(p) = \frac{n}{pq}$ (for fixed n)

Q1- Tossing a coin

No. of trials, $n = 5$

$$P(H) = 0.5 \quad P(T) = 0.5$$

What is prob. of getting exactly 3 heads in 5 flips?

$$n = 5$$

$$p = 0.5 \quad q = 0.5$$

$$k = 3$$

$$P(X=3) = {}^5C_3 (0.5)^3 (1-0.5)^{5-3} = 0.31$$

Q2-

Inspecting 10 items in a factory. where each item has a 10% chance of being defective.

No. of trials (n) = 10

Prob. of success (p) = 0.1 (defective item)

No. of success (k) = varies from 0 to 10

What is the probability of finding exactly 2 defective items in a sample of 10?

$$P(X=2) = {}^{10}C_2 (0.1)^2 (0.9)^8 \\ = 0.193$$

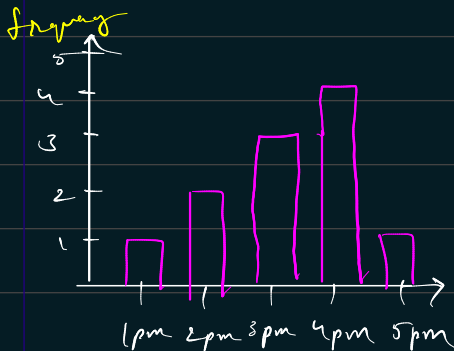
③ Poisson Dist.

-> is a discrete probability distribution that expresses the probability of a given number of events occurring in a fixed interval of time if these events occur with a known constant mean rate and independently of the time since the last event.

Discrete $\rightarrow$ PMF

Ex: No. of people visiting hospital every hour

fixed interval of time



$\lambda = 3 \Rightarrow$ Expected no. of events occurring at every time interval (min)

PMF:

$$P(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$P(x=5) = \frac{e^{-3} 3^5}{5!} \approx 0.101 \approx 10.1\%$$

Probability of a person visiting at 5pm

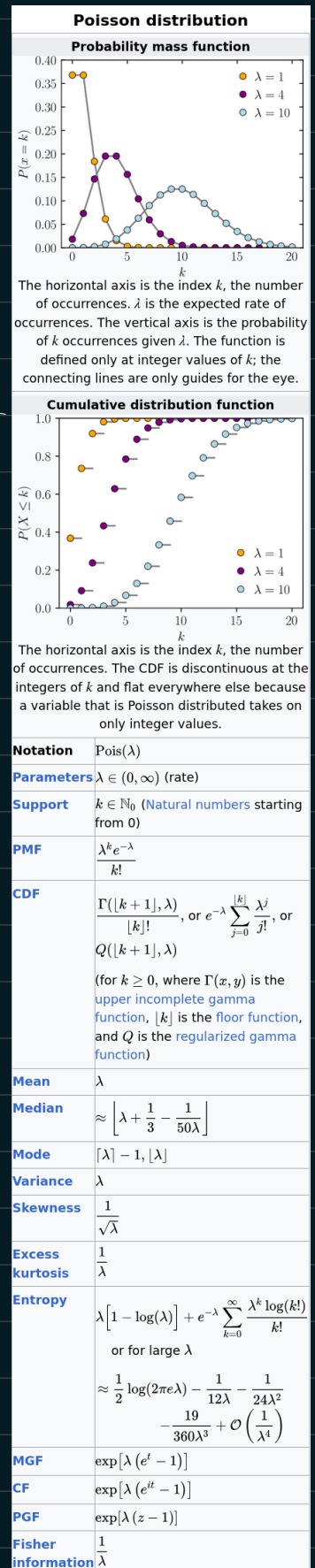
Mean =

$$E(x) = \mu = \lambda \cdot t$$

$t \rightarrow$ time interval

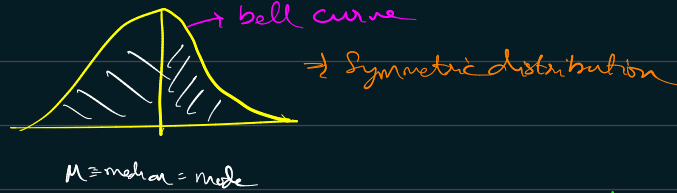
Variance :-

$$\lambda$$



⑨ Normal / Gaussian Dist.

-> type of continuous probability distribution for a real-valued random variable



Notation:

$$N(\mu, \sigma^2)$$

Parameters:

$\mu \in \mathbb{R}$ = mean

$\sigma^2 \in \mathbb{R}_{>0}$ = Variance

$x \in \mathbb{R}$ → Random Variable

Ex: Heights of students

IRIS Dataset → petal length, sepal length
petal width, sepal width

$$\text{PDF} = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

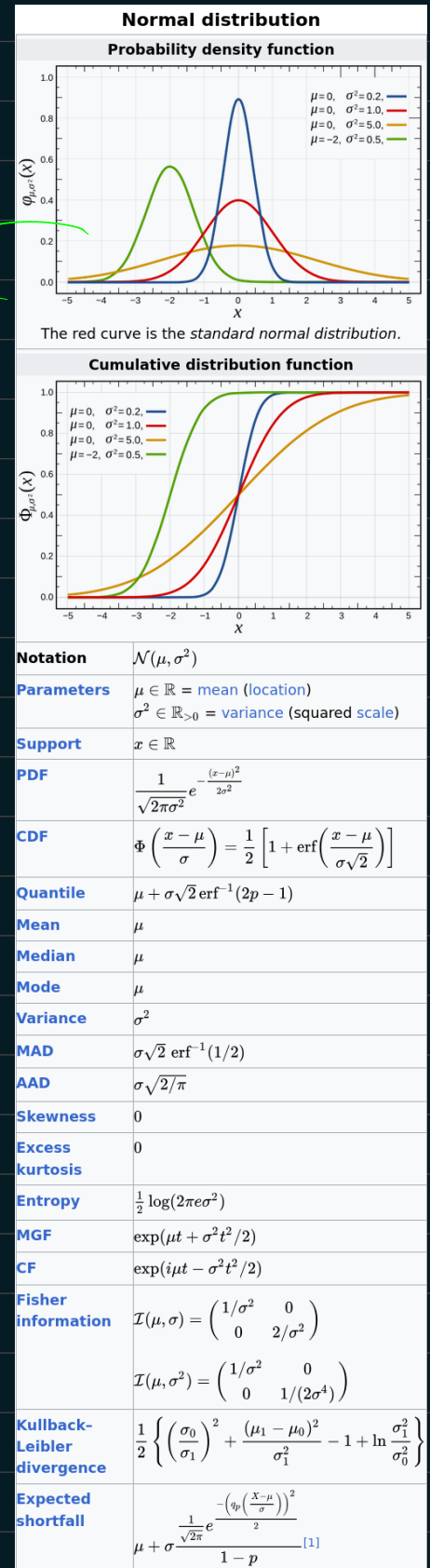
Mean

$$\mu = \sum_{i=1}^n \frac{x_i}{n}$$

Variance

$$\sigma^2 = \sum_{i=1}^n \frac{(x_i - \mu)^2}{n}$$

$\sigma^2 \uparrow \uparrow \uparrow$
Spread ↑↑



Empirical rule for Gaussian dist:-

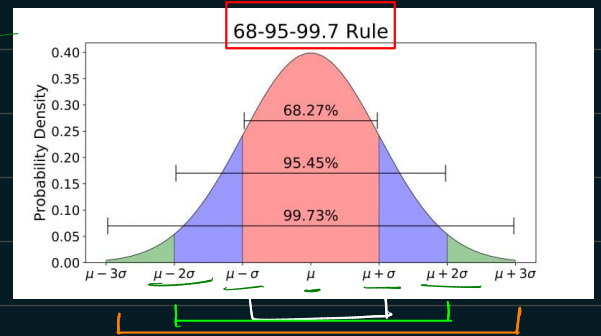
$X = [\text{-----}] \Rightarrow \text{normal / Gaussian dist.}$

$\Rightarrow$

$$P(\mu - \sigma \leq X \leq \mu + \sigma) \approx 68\%$$

$$P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 95\%$$

$$P(\mu - 3\sigma \leq X \leq \mu + 3\sigma) \approx 99\%$$

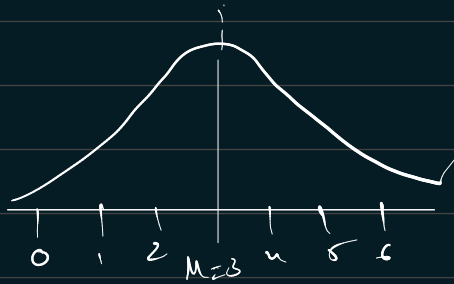


$\rightarrow$ To check if a dist is normal dist $\rightarrow$ use Q-Q plot

⑤ Standard normal distribution & Z score

$$X = \{1, 2, 3, 4, 5\} \quad \mu = 3$$

$$\sigma = 1.414 \approx 1$$



Convert it
to have

$$\mu = 0$$

$$\sigma = 1$$

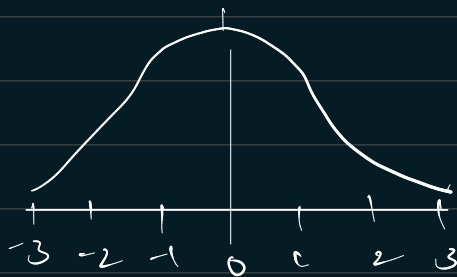
using Z score formula

$$Z \text{ score} = \frac{x_i - \mu}{\sigma}$$

$$① \quad \frac{1-3}{1} = -2$$

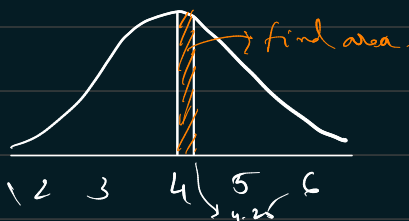
$$② \quad \frac{2-3}{1} = -1$$

$$y = \{-2, -1, 0, 1, 2\}$$



Standard normal dist $\rightarrow \mu = 0, \sigma = 1$

Ex: How many sd 4.25 is away from the mean



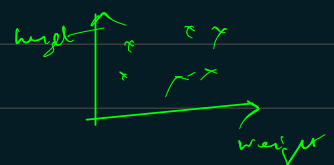
$$\mu = 4$$

$$\sigma = 1$$

$$x_i = 4.25$$

$$Z \text{ score} = \frac{4.25 - 4}{1} = 0.25$$

$\rightarrow$ Z score is used for standardization of the data, if not the datapoints across various data gets scattered



⑥ Uniform Distribution:-

a) Continuous Uniform dist (PDF)

b) Discrete Uniform dist (PMF)

a) Continuous Uniform dist.

-> the continuous uniform distributions or rectangular distributions are a family of symmetric probability distributions.

-> Such a distribution describes an experiment where there is an arbitrary outcome that lies between certain bounds.

-> The bounds are defined by the parameters, a and b which are the minimum and maximum values.

-> The interval can either be closed (i.e. $[a, b]$) or open (i.e. (a, b))

Notation:-

$$U(a, b)$$

Parameters:-

$$-\infty < a < b < \infty$$

$$PDF = \begin{cases} \frac{1}{b-a} & x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$$

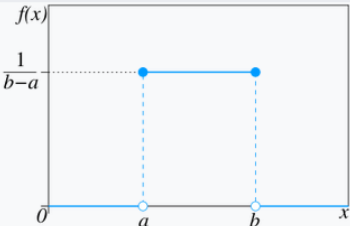
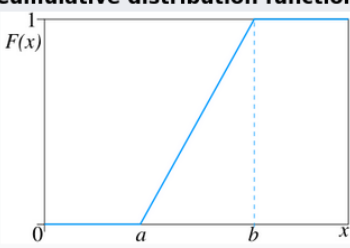
$$CDF = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & x \in [a, b] \\ 1 & x > b \end{cases}$$

Mean:-

$$\frac{1}{2}(a+b)$$

Median:-

$$\frac{1}{2}(a+b)$$

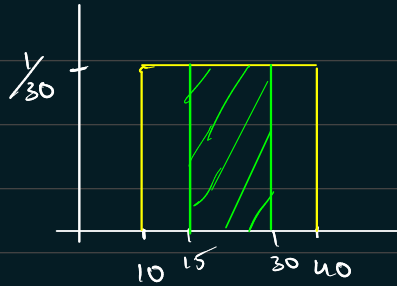
Continuous uniform	
Probability density function  Using maximum convention	
Cumulative distribution function 	
Notation	$\mathcal{U}_{[a,b]}$
Parameters	$-\infty < a < b < \infty$
Support	$[a, b]$
PDF	$\begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$
CDF	$\begin{cases} 0 & \text{for } x < a \\ \frac{x-a}{b-a} & \text{for } x \in [a, b] \\ 1 & \text{for } x > b \end{cases}$
Mean	$\frac{1}{2}(a+b)$
Median	$\frac{1}{2}(a+b)$
Mode	any value in (a, b)
Variance	$\frac{1}{12}(b-a)^2$
MAD	$\frac{1}{4}(b-a)$
Skewness	0
Excess kurtosis	$-\frac{6}{5}$
Entropy	$\log(b-a)$
MGF	$\begin{cases} \frac{e^{ib} - e^{ia}}{i(b-a)} & \text{for } t \neq 0 \\ 1 & \text{for } t = 0 \end{cases}$
CF	$\begin{cases} \frac{e^{itb} - e^{ita}}{it(b-a)} & \text{for } t \neq 0 \\ 1 & \text{for } t = 0 \end{cases}$

Variance:

$$\frac{1}{12} (b-a)$$

Ex: The no. of candies sold daily at a shop is uniformly distributed with a max of 40 & min of 10.

i: Prob of daily sales to fall b/w 15 & 30?



$$\frac{1}{b-a} = \frac{1}{40-10} = \frac{1}{30}$$

$$x_1 = 15$$

$$x_2 = 30$$

$$P(15 \leq x \leq 30) = (x_2 - x_1) \cdot \frac{1}{b-a}$$

$$= (30 - 15) \cdot \frac{1}{30}$$

$$= \frac{1}{2}$$

$$P(x \geq 20) = (40 - 20) \cdot \frac{1}{20}$$

$$= \frac{2}{3}$$

b) Discrete Uniform dist:-

-> discrete uniform distribution is a symmetric probability distribution wherein each of some finite whole number n of outcome values are equally likely to be observed.

-> Thus every one of the n outcome values has equal probability $1/n$. Intuitively, a discrete uniform distribution is "a known, finite number of outcomes all equally likely to happen."

discrete random variable $\rightarrow$ PMF

Ex:- Rolling a dice:-

$$P(1) = P(2) = P(3) = \dots = 1/6$$

$$x = \{ \underset{a}{1}, \underset{2}{2}, \underset{3}{3}, \underset{4}{4}, \underset{5}{5}, \underset{b}{6} \}$$

$$\frac{1}{n} \Rightarrow n = b - a + 1$$

$$\Rightarrow 6 - 1 + 1 = 6$$

Notation:-

$$U(a, b)$$

Parameters:-

$$a, b \quad ; \quad b \geq a$$

$$\text{PMF} = 1/n$$

Mean:-

$$\frac{a+b}{2}$$

Discrete uniform	
Probability mass function	
Cumulative distribution function	
Notation	$U\{a, b\}$ or $\text{unif}\{a, b\}$
Parameters	a, b integers with $b \geq a$ $n = b - a + 1$
Support	$k \in \{a, a+1, \dots, b-1, b\}$
PMF	$\frac{1}{n}$
CDF	$\frac{ k - a + 1}{n}$
Mean	$\frac{a+b}{2}$
Median	$\frac{a+b}{2}$
Mode	N/A
Variance	$\frac{(b-a+1)^2 - 1}{12}$
Skewness	0
Excess kurtosis	$\frac{6(n^2 + 1)}{5(n^2 - 1)}$
Entropy	$\ln(n)$
MGF	$\frac{e^{at} - e^{(b+1)t}}{n(1 - e^t)}$
CF	$\frac{e^{iat} - e^{i(b+1)t}}{n(1 - e^{it})}$
PGF	$\frac{z^a - z^{b+1}}{n(1 - z)}$

⑦ log normal dist.

-> Log-normal distribution is a continuous probability distribution of a random variable whose logarithm is normally distributed.

-> Thus, if the random variable X is log-normally distributed, then $Y = \ln(X)$ has a normal distribution. Equivalently, if Y has a normal distribution, then the exponential function of Y , $X = \exp(Y)$, has a log-normal distribution.

$X \rightarrow \text{log normal dist } (\mu, \sigma)$

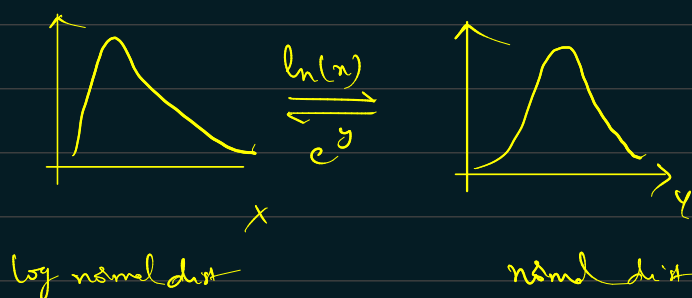
$\downarrow (\text{transformation})$

$Y = \ln(X) \rightarrow \text{normal dist.}$

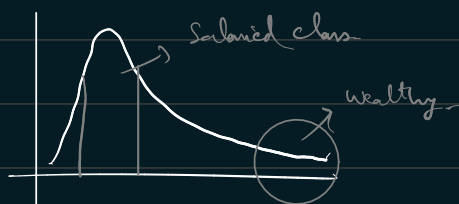
$\updownarrow$

* if $Y \rightarrow \text{normal dist.}$

then $X = e^Y \rightarrow \text{log normal dist.}$



Go-wealth dist. of world



length of comments in discussion forum

length of chess game.

Log-normal distribution	
Probability density function	
Identical parameter μ but differing parameters σ	
Cumulative distribution function	
$\mu = 0$	
Notation	$\text{Lognormal}(\mu, \sigma^2)$
Parameters	$\mu \in (-\infty, +\infty)$ (logarithm of location), $\sigma > 0$ (logarithm of scale)
Support	$x \in (0, +\infty)$
PDF	$\frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right)$
CDF	$\frac{1}{2} \left[1 + \text{erf}\left(\frac{\ln x - \mu}{\sigma\sqrt{2}}\right) \right] = \Phi\left(\frac{\ln(x) - \mu}{\sigma}\right)$
Quantile	$\exp\left(\mu + \sqrt{2\sigma^2} \text{erf}^{-1}(2p - 1)\right)$ $= \exp(\mu + \sigma\Phi^{-1}(p))$
Mean	$\exp\left(\mu + \frac{\sigma^2}{2}\right)$
Median	$\exp(\mu)$
Mode	$\exp(\mu - \sigma^2)$
Variance	$[\exp(\sigma^2) - 1] \exp(2\mu + \sigma^2)$
Skewness	$[\exp(\sigma^2) + 2] \sqrt{\exp(\sigma^2) - 1}$
Excess kurtosis	$\exp(4\sigma^2) + 2 \exp(3\sigma^2) + 3 \exp(2\sigma^2) - 6$
Entropy	$\log_2\left(\sqrt{2\pi} \sigma e^{\mu + \frac{1}{2}}\right)$
MGF	defined only for numbers with a non-positive real part, see text
CF	representation $\sum_{n=0}^{\infty} \frac{(it)^n}{n!} e^{n\mu + n^2\sigma^2/2}$ is asymptotically divergent, but adequate for most numerical purposes
Fisher information	$\begin{pmatrix} -\frac{1}{\sigma^2} & 0 \\ 0 & \frac{2}{\sigma^2} \end{pmatrix}$
Method of moments	$\mu = \log\left(\frac{\mathbb{E}[X]}{\sqrt{\frac{\text{Var}[X]}{\mathbb{E}[X]^2} + 1}}\right),$ $\sigma = \sqrt{\log\left(\frac{\text{Var}[X]}{\mathbb{E}[X]^2} + 1\right)}$
Expected shortfall	$\frac{1}{2} e^{\mu + \frac{\sigma^2}{2}} \frac{1 + \text{erf}\left(\frac{\sigma}{\sqrt{2}} + \text{erf}^{-1}(2p - 1)\right)}{p}$ [1] $= e^{\mu + \frac{\sigma^2}{2}} \frac{1}{1-p} (1 - \Phi(\Phi^{-1}(p) - \sigma))$

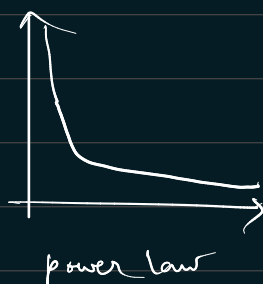
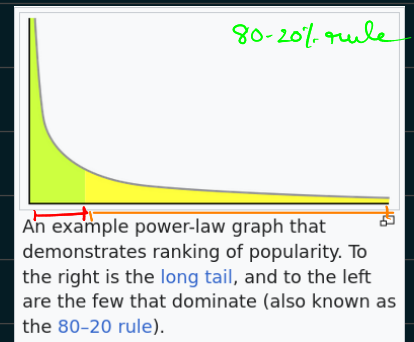
⑧ Power law dist.:-

- > a power law is a functional relationship between two quantities, where a relative change in one quantity results in a relative change in the other quantity proportional to the change raised to a constant exponent: one quantity varies as a power of another.
- > The change is independent of the initial size of those quantities.

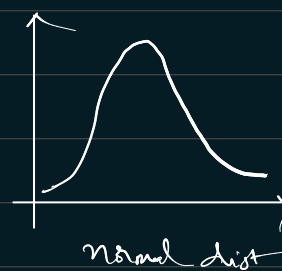
Ex: 80% wealth is dist. with 20% pop-

→ 20% teams wins 80% of IPL matches

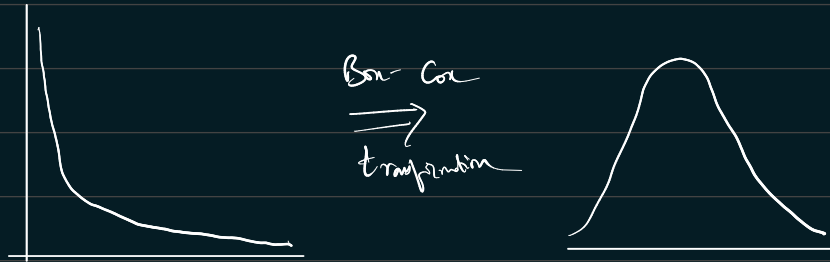
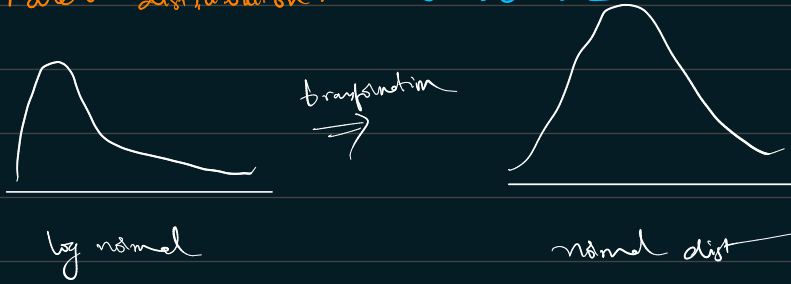
→ frequency of words used in languages



Box-Cox
⇒
transformation

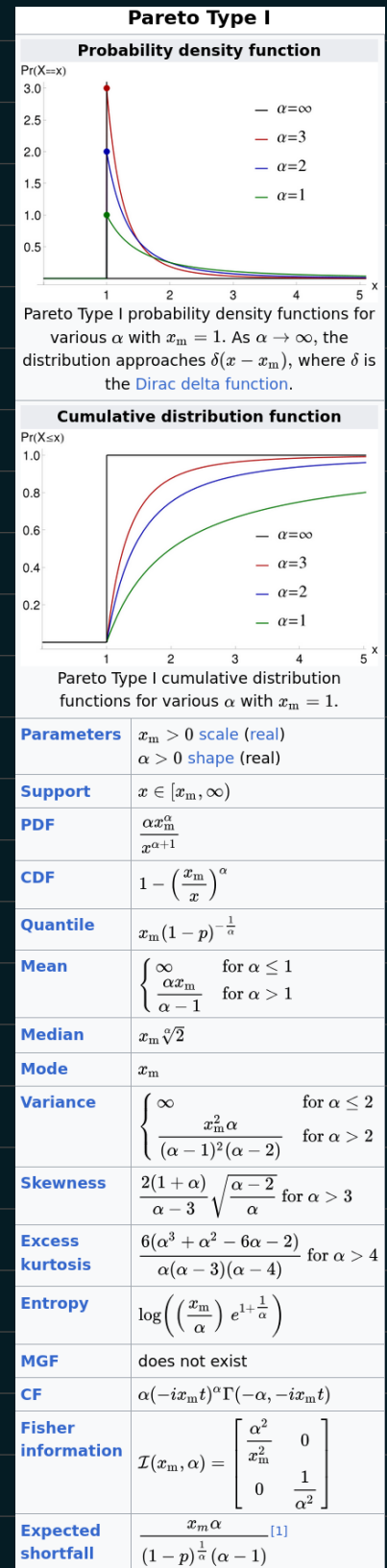


⑨ Pareto distribution: 80-20 rule



Ex- 80% of defects can be solved if we save 20% of defects.

⇒ Pareto dist. follows Power law.

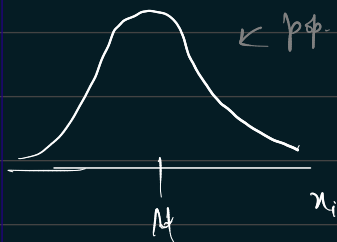


Central limit theorem:-

-> Central limit theorem relies on the concept of a sampling distribution, which is the probability dist. of a statistic for a large number of samples taken from a population.

-> The central limit theorem says that the sampling dist. of the mean will always be normal distributed, as long as the sample size is large enough. Regardless of whether the population has a normal, Poisson, binomial or any other distribution, the sampling dist. of the mean will be normal.

$$a) X \sim N(\mu, \sigma)$$

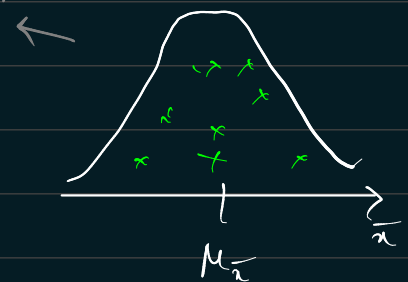


← pop. data follows normal dist.

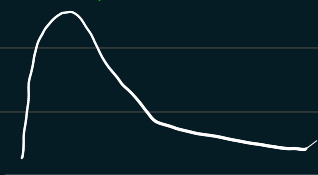
$$\begin{aligned} S_1 &= \{x_1, x_2, \dots, x_n\} \Rightarrow \bar{x}_1 \\ S_2 &= \{x_1, x_2, x_3, \dots, x_n\} \Rightarrow \bar{x}_2 \\ S_3 &= \{x_1, x_2, x_3, \dots, x_n\} \Rightarrow \bar{x}_3 \\ &\vdots \\ S_m &= \dots \Rightarrow \bar{x}_m \end{aligned}$$



normal dist. ←



$$b) X \not\sim N(\mu, \sigma)$$



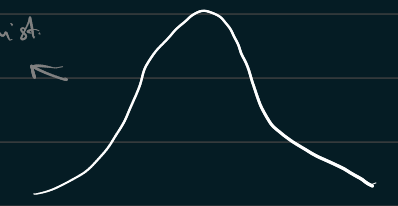
$$n \geq 30$$

↓
for non Gaussian
dist.

$$\begin{aligned} S_1 &\rightarrow \bar{x}_1 \\ S_2 &\rightarrow \bar{x}_2 \\ &\vdots \\ S_m &\rightarrow \bar{x}_m \end{aligned}$$



normal dist. ←



Estimates

-> Specified observed numerical value used to estimate an unknown population parameter.

① Point Estimate :-

Single numerical value used to estimate an unknown p.p. parameter

Ex:- Sample mean is a point estimate of p.p. mean



② Interval Estimate :-

Range of values is used to estimate unknown p.p. parameter

